UNIVERSITY NAME

Rigid Body Mechanics — Final Exam

Year: 2014

Exercise 1:

Let $R_0(O, \bar{x}_0, \bar{y}_0, \bar{z}_0)$ be a Galilean orthonormal direct frame relative to which we propose to study the motion of a system (Σ) consisting of:

- A straight rod (T) along the direction class$(O, \bar{x})$, with negligible cross-section and mass, which can rotate freely around its fixed endpoint O. - A solid homogeneous sphere (S) of mass M, radius a, and center G.

The sphere (S) is pierced along its axis $(G, \bar{x})$ by the rod (T). It can slide along the rod and rotate around it.

All friction is neglected (i.e., the different contacts are considered frictionless) and we assume the connection at O is perfect. The motion parameters of (Σ) in the frame R_0 are the three usual Euler angles (ψ, θ, ϕ) and $\overline{OG} = \rho(t)$.

All results will be expressed in the basis $(i, \bar{w}, \bar{z})$.

1. Determine the following instantaneous rotation vectors:

$$\hat{\Omega}(T/R_0), \quad \hat{\Omega}(S/T), \quad \hat{\Omega}(S/R_0)$$

2. Calculate the velocity and acceleration vectors of point G relative to R_0.

3. Calculate the inertia matrix at point O of the system Σ:

$$\Pi_0(\Sigma)$$

4. Calculate the angular momentum about O of the system Σ relative to R_0:

$$\tilde{L}_0(\Sigma/R_0)$$

5. Show that:

$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$

   Deduce two equations of motion in the form of first integrals.

6. Determine the kinetic energy of the system Σ relative to the frame R_0.

7. By applying the kinetic energy theorem to Σ, give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)

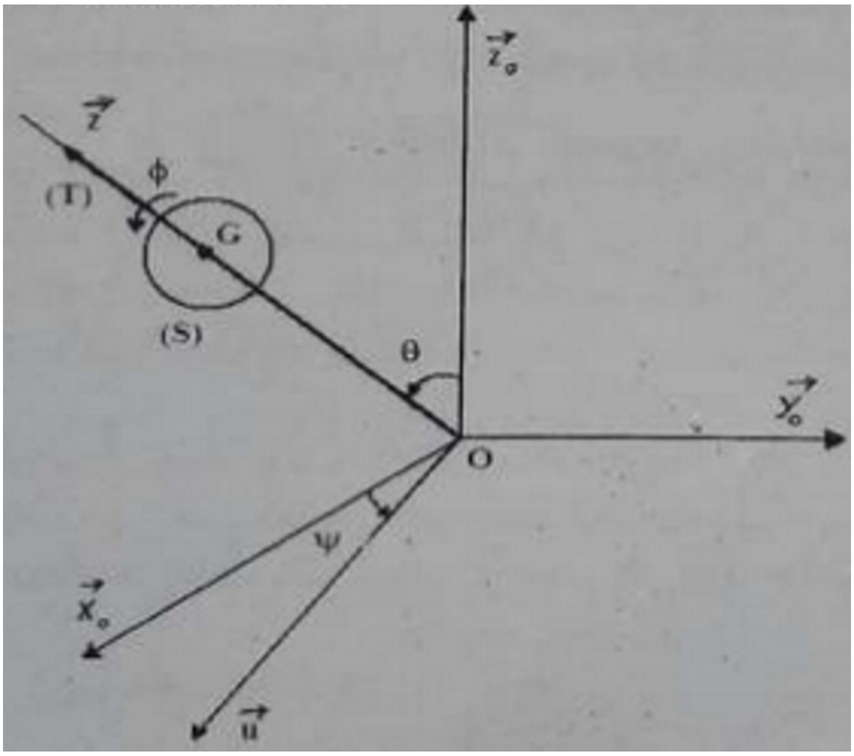
8. In the special case where the axis $(O, \bar{z})$ remains coincident with the axis $(O, \bar{z}_0)$, determine $\rho(t)$. Given:
$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$
Show that:
$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$
Deduce two equations of motion in the form of first integrals.

9. Determine the kinetic energy of the system Σ relative to the frame R_0.

10. By applying the kinetic energy theorem to Σ, give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)

11. In the special case where the axis $(O, \bar{z})$ remains coincident with the axis $(O, \bar{z}_0)$, determine $\rho(t)$. Given:
$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$

System Diagram



Answer Area

1. Rotation vectors:

$$\begin{aligned}\hat{\Omega}(T/R_0) &= \dot{\psi}\,\bar{z}_0 + \dot{\theta}\,\bar{u}\\ \hat{\Omega}(S/T) &= \dot{\phi}\,\bar{x}\\ \hat{\Omega}(S/R_0) &= \hat{\Omega}(T/R_0) + \hat{\Omega}(S/T)\\ &= \dot{\psi}\,\bar{z}_0 + \dot{\theta}\,\bar{u} + \dot{\phi}\,\bar{x}\end{aligned}$$

where $\bar{u} = \bar{z}_0 \times \bar{x}/\|\bar{z}_0 \times \bar{x}\|$ in the nodal line direction.

2. Velocity and acceleration of G:

$$\begin{aligned}\bar{V}(G/R_0) &= \dot{\rho}\,\bar{x} + \rho\,\hat{\Omega}(T/R_0) \times \bar{x}\\ &= \dot{\rho}\,\bar{x} + \rho\dot{\theta}\,\bar{u} \times \bar{x} + \rho\dot{\psi}\sin\theta\,\bar{y}\\ \bar{a}(G/R_0) &= \frac{d}{dt}\bar{V}(G/R_0)\bigg|_{R_0}\end{aligned}$$

(Full expansion omitted for brevity, follows from differentiation of velocity.)

3. Inertia matrix at O of Σ:
Rod (T) mass negligible, only sphere contributes.
Parallel axis theorem:

$$\Pi_O(\Sigma) = \Pi_G(S) + M\begin{pmatrix}\rho^2 & 0 & 0\\ 0 & \rho^2 + \frac{2}{5}a^2 & 0\\ 0 & 0 & \rho^2 + \frac{2}{5}a^2\end{pmatrix}$$

in basis $(\bar{x}, \bar{w}, \bar{z})$ with $\bar{w}$ perpendicular to $\bar{x}$ in the plane of $\bar{x}$ and $\bar{z}_0$.

4. Angular momentum about O:

$$\begin{aligned}\tilde{L}_O(\Sigma/R_0) &= \tilde{L}_O(S/R_0)\\ &= \mathbf{I}_O(S)\cdot\hat{\Omega}(S/R_0) + M\,\overline{OG} \times \bar{V}(G/R_0)\end{aligned}$$

where $\mathbf{I}_O(S)$ is the inertia tensor of sphere relative to O.

5. Conservation laws:
No external torque about $\bar{z}_0$ and $\bar{z}$ axes $\Rightarrow$

$$\tilde{L}_O(\Sigma/R_0)\cdot\bar{z}_0 = k_1, \quad \tilde{L}_O(\Sigma/R_0)\cdot\bar{z} = k_2$$

These give two first integrals of motion (conservation of angular momentum components along $\bar{z}_0$ and $\bar{z}$).

6. Kinetic energy:

$$T = \frac{1}{2}M\bar{V}^2(G/R_0) + \frac{1}{2}\hat{\Omega}(S/R_0)\cdot\mathbf{I}_G(S)\cdot\hat{\Omega}(S/R_0)$$

with $\mathbf{I}_G(S) = \frac{2}{5}Ma^2\,\mathbf{1}$ (isotropic).

7. Kinetic energy theorem:
No non-conservative work (perfect connections, no friction) $\Rightarrow$

$$\frac{dT}{dt} = 0 \quad \text{or} \quad T = \text{const} = E_0$$

which is a third first integral (conservation of total mechanical energy if no potential).

8. Special case $\bar{z} = \bar{z}_0$:
Then $\theta = 0$, $\bar{u}$ undefined, rotation simplifies:

$$\hat{\Omega}(S/R_0) = \dot{\psi}\bar{z}_0 + \dot{\phi}\bar{x}$$

Conservation laws give:

$$\frac{2}{5}Ma^2(\dot{\psi} + \dot{\phi}) + M\rho^2\dot{\psi} = k_1$$

$$\frac{2}{5}Ma^2\dot{\phi} = k_2$$

Energy conservation:

$$\frac{1}{2}M\dot{\rho}^2 + \frac{1}{2}\cdot\frac{2}{5}Ma^2(\dot{\psi}^2 + \dot{\phi}^2 + 2\dot{\psi}\dot{\phi}) + \frac{1}{2}M\rho^2\dot{\psi}^2 = E_0$$

System reduces to ODE for $\rho(t)$ using constants k_1, k_2, E_0.

11. Determination of $\rho(t)$ in special case:
From energy and angular momentum conservation, after eliminating $\dot{\psi}, \dot{\phi}$, equation for ρ is:

$$\frac{1}{2}M\dot{\rho}^2 + \frac{k_2^2}{2\cdot\frac{2}{5}Ma^2} + \frac{(k_1 - k_2)^2}{2M\rho^2} = E_0$$

This resembles a 1D central motion:

$$\frac{1}{2}M\dot{\rho}^2 + V_{\text{eff}}(\rho) = E_0$$

with effective potential

$$V_{\text{eff}}(\rho) = \frac{A}{\rho^2} + B, \quad A, B \text{ constants.}$$

Solving with $\rho(0) = 0, \dot{\rho}(0) = \rho_0$ yields oscillatory or monotonic motion depending on constants.